# Hybrid Semantic Context Forests and Kyphosis-Aware Deformable Topology Matching for Automatic 3D Rib Centerline Extraction and Labeling in Thoracic CT

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**Abstract.** Automatic extraction and anatomical numbering (R01–R12, L01–L12) of all 24 rib centerlines in thoracic CT is a prerequisite for unfolded rib visualisation, fracture detection and surgical navigation, but remains difficult because of adjacent-bone confluence (clavicle, scapula, sternum, vertebrae), calcified costal cartilage bridges, kyphotic pose variation and the frequent partial visibility of the first and twelfth ribs. We present ribline, an open-source three-stage framework that combines (i) an automatically detected lung-apex / vertebral-midline coordinate frame with a kyphosis-aware azimuthal angle used to order points along each rib, (ii) a hybrid voxel detector that max-fuses an anatomically anchored Semantic Context Forest (SCF) with a lightweight 3D U-Net, and (iii) a global labeling stage that decomposes the probability skeleton into candidate curves, assigns rib indices with a patient-scaled bilateral Viterbi dynamic program coupled across the two hemithoraces, and refines terminal ribs with template recovery and normal-plane centring (RibSnake). On the 60-scan Siemens Corporate Research (SCR) 2 mm thoracic CT benchmark (47 training / 12 evaluation scans), using a symmetric 10 mm labeling criterion and a 50 mm penalty for missed ribs, the full pipeline reaches a labeled symmetric Chamfer distance of $2.05 \pm 0.48$ mm (95 % bootstrap CI $[1.80, 2.33]$), labeled clDice of 0.886 (3 mm) / 0.945 (4 mm), 97.2 % labeling accuracy (278/286 ribs, Wilson 95 % CI 94.6–98.6 %), 97.2 % recall and no false-positive curves. We report every metric with confidence intervals, make the metric definitions, training recipes and checkpoints consistent, and state explicitly that the Stage-3 constants were developed with access to the 12 evaluation scans. Applied unchanged to the 47 training scans (1,116 ribs), on which those constants were never inspected, the frozen pipeline reaches $2.63 \pm 1.91$ mm (median 1.89 mm) and 95.4 % labeling accuracy; in a 5-fold cross-validation over all 59 scans (1,402 ribs) with per-fold refitted atlas, SCF and 3D U-Net and frozen Stage-3 constants, it reaches $3.64 \pm 3.90$ mm (95 % CI $[2.79, 4.74]$, median 2.39 mm) and 91.8 % labeling accuracy ($p < 10^{-5}$ over standalone detectors and $p = 2 \times 10^{-9}$ for RibSnake refinement).

**Code:** https://github.com/wq2012/ribline
**Pretrained Models:** https://huggingface.co/collections/wq2012/ribline

**Keywords:** Rib centerline extraction, anatomical rib labeling, semantic context forests, 3D U-Net, Viterbi dynamic programming, deformable template matching, thoracic CT

## 1 Introduction

The human rib cage comprises twelve pairs of curved, tubular bones (`R01`–`R12`, `L01`–`L12`). Accurate 3D extraction and individual numbering of rib centerlines in CT enables unfolded rib cage visualisation via curved planar reformation for trauma screening [9,10], provides an anatomical coordinate system for localising thoracic lesions [5], supports radiotherapy and spinal instrumentation planning [6], and is the topological backbone of automated rib-fracture detection [4].

Despite the high attenuation of cortical bone, fully automatic 3D rib centerline tracing and numbering is hard for four reasons: (1) *adjacent bone confluence*: the first rib articulates with the manubrium and lies next to the clavicle and scapula, so skeletonisation leaks into the shoulder girdle; (2) *intercostal ossification*: calcified costal cartilage and partial-volume averaging create bridges between ribs 5–10, producing false junctions and hairpin U-turns; (3) *terminal-rib variability*: the floating twelfth ribs vary widely in length and are congenitally absent in a few percent of patients, so naive top-down or bottom-up counting suffers index shifts whenever a terminal rib is missed; and (4) *thoracic kyphosis*: in the upper thorax the rib cage curves anteriorly and inferiorly while the spine arches posteriorly, so a polar angle around the torso centroid is not a reliable ordering parameter along the posterior rib neck.

Early methods relied on seeded region growing or ridge tracking [8,5,6,7]. Wu *et al.* [1] introduced learning-based deformable template matching with Marginal Space Learning [12] on the SCR data set used here, reporting 89.5–95.9 % labeling accuracy under their own protocol. Auto-context [13] and Semantic Context Forests (SCF) [2,3] showed that combining multi-scale differential features with long-range random-shift intensity/probability differences gives discriminative voxel classifiers for thin structures. Deep 3D networks [11,10] and point-cloud networks such as RibSeg v2 [4] achieve strong segmentation, but on small training cohorts standalone CNNs still under-segment the small first and floating twelfth ribs and need post-hoc heuristics for numbering.

**Contributions.** We describe and release (Fig. 1) a three-stage pipeline whose methods section below is written to match the released code line by line:

1. a **lung-apex / vertebral-midline thoracic frame** and a **kyphosis-aware azimuthal angle** $\theta(\mathbf{x})$ whose focus moves posteriorly towards the apex; on the 1,402 annotated ribs of the 59 valid SCR scans $\theta$ increases from the posterior to the anterior end of *every* rib and is strictly monotonic on 78.6 % of ribs (vs. 57.0 % for a torso-centre angle; Sec. 2);
2. a **complementary hybrid detector**: an anatomically anchored SCF (better on first ribs) max-fused with a plain 3D U-Net (better on intermediate ribs);

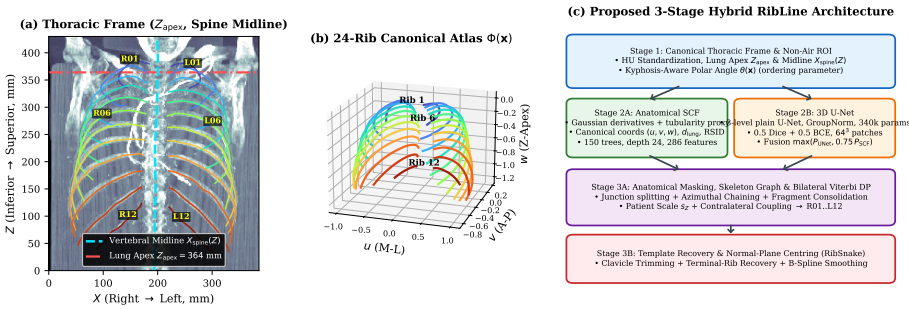



**Fig. 1.** Overview of `ribline`. (a) Coronal MIP with the detected lung apex $Z_{\text{apex}}$, the vertebral midline $X_{\text{spine}}(Z)$ and the ground-truth centerlines. (b) The 24-rib canonical atlas $\bar{\mathcal{C}}_{s,k}$ fitted on the 47 training scans. (c) Flowchart of the hybrid SCF + 3D U-Net detector, the bilateral Viterbi labeler and the `RibSnake` refinement.

3. a **patient-scaled bilateral Viterbi labeler** with contralateral coupling and outlier tolerance, plus `RibSnake` (clavicle trimming, terminal-rib template recovery, normal-plane centring);
4. a **corrected evaluation protocol** (symmetric labeling criterion, missed-rib penalty, precision, bootstrap and Wilson confidence intervals, paired tests), a transparent treatment of one annotation error in the public ground truth, and a cross-validation with frozen Stage-3 constants that quantifies how much the development protocol favoured the 12 evaluation scans.

## 2 Canonical Thoracic Frame and Statistical Rib Atlas

### 2.1 Hounsfield Standardisation and Thoracic Landmark Detection

The SCR archive mixes signed Hounsfield units (air $\approx -1000$) and unsigned shifted values (air $\approx 0$). If the minimum intensity of a volume is $\geq -100$, we subtract 1024; otherwise the volume is used as is (`standardize_hu`).

All landmarks are computed on a 2× down-sampled grid. The torso mask is $I > -350\,\text{HU}$ with slice-wise hole filling; internal air is the filled torso minus voxels with $I \geq -380\,\text{HU}$, morphologically opened and reduced to the (at most four) largest components with more than 800 sub-voxels, which form the lung mask $\mathcal{M}_{\text{lung}}$. The **lung apex** $Z_{\text{apex}}$ is the larger of the per-side 99.5th-percentile $Z$ coordinates of $\mathcal{M}_{\text{lung}}$. The **spine** is located as the peak of a Gaussian-smoothed ($\sigma = 3$ sub-voxels) axial projection of bone intensity ($\text{clip}(I - 120, 0, 600)$) restricted to the posterior 40 % of the antero-posterior bone extent and to $\pm 50\,\text{mm}$ around the medio-lateral bone centre, refined by an intensity-weighted centroid; a per-slice median of bone voxels within $\pm 32 \times \pm 36\,\text{mm}$ of this seed, smoothed with a Gaussian of $\sigma = 12$ slices, yields the trajectory $(X_{\text{spine}}(Z), Y_{\text{spine}}(Z))$; the scalar seed values $(x_{\text{spine}}, y_{\text{spine}})$ are also kept. The thoracic half-width $W$ is the 96th percentile of $|X - x_{\text{spine}}|$ and $Y_{\text{ant}}, Y_{\text{post}}$ the 5th/95th percentiles of $Y$

over bone voxels ($I > 150\,\mathrm{HU}$) in the band $Z_{\mathrm{apex}} - 210 \le Z \le Z_{\mathrm{apex}} - 10\,\mathrm{mm}$; $D = \max(Y_{\mathrm{post}} - Y_{\mathrm{ant}}, 100\,\mathrm{mm})$.

Measured on the 59 valid scans (script `polar_angle_stats`), the posterior (spine-side) end of `R01`/`L01` lies $+11.9 \pm 9.3\,\mathrm{mm}$ above $Z_{\mathrm{apex}}$ ($n = 117$; median 13.0 mm, 5–95 % range $-2.5$ to 24.5 mm), and the midline error $|X_{\mathrm{spine}}(Z) - \frac{1}{2}(X^{\mathrm{R}}_{\mathrm{post}} + X^{\mathrm{L}}_{\mathrm{post}})|$ over 695 annotated rib pairs is $2.5 \pm 2.4\,\mathrm{mm}$ (median 1.8 mm). The apex is therefore a useful but not a tight anchor for the first rib; the labeler below never relies on it alone.

### 2.2 Canonical Mapping and Kyphosis-Aware Azimuthal Angle

A physical point $\mathbf{x} = (X, Y, Z)$ is mapped to canonical coordinates

$$u = \frac{X - x_{\mathrm{spine}}}{W}, \qquad v = \frac{Y - y_{\mathrm{spine}}}{D}, \qquad w = \frac{Z - Z_{\mathrm{apex}}}{240\ \mathrm{mm}}, \tag{1}$$

using the scalar spine seed (the per-slice trajectory is used only for the anatomical masks of Sec. 4). To order points along a rib from its vertebral head to its sternal end we use an azimuthal angle whose focus sits $x_c = 8\,\mathrm{mm}$ lateral of the midline and moves posteriorly towards the apex,

$$y_c(Z) = Y_{\mathrm{ant}} + 0.25\,D + 0.16\,\mathrm{clip}(Z_{\mathrm{apex}} - Z,\ 0,\ 220\ \mathrm{mm}), \tag{2}$$

$$\theta(\mathbf{x}) = \mathrm{atan2}\big(-(Y - y_c(Z)),\ |X - x_{\mathrm{spine}}| - x_c\big), \tag{3}$$

followed by phase unwrapping along the curve. Over the 1,402 ground-truth ribs, $\theta_{\mathrm{end}} > \theta_{\mathrm{start}}$ (correct global orientation) holds for 100 % of ribs; $\theta$ is *strictly* monotonic for 1,102 ribs (78.6 %), the largest local reversal exceeds 0.1 rad on 8 ribs (0.6 %) and 0.3 rad on 3 ribs. The naive angle $\mathrm{atan2}(-(Y - y_{\mathrm{lung}}), |X - x_{\mathrm{spine}}|)$ around a fixed torso centre is strictly monotonic for only 799 ribs (57.0 %) with reversals $> 0.1$ rad on 107 ribs (7.6 %). All downstream uses of $\theta$ (orientation, fragment chaining, angular span) are robust to small reversals.

### 2.3 Non-Parametric 24-Rib Statistical Atlas

From the 47 training scans (1,116 ribs) each rib is oriented posterior-to-anterior, resampled to $M = 80$ arc-length-uniform points, mapped by Eq. 1 and averaged to a mean curve $\bar{\mathcal{C}}_{s,k} \in \mathbb{R}^{80\times 3}$ per side $s$ and index $k$ (Fig. 1b). For every $k$ we also store mean and standard deviation $(\boldsymbol{\mu}_{f,k}, \boldsymbol{\sigma}_{f,k})$ of the 5-D descriptor $\mathbf{f}(\mathcal{C}) = [z_{\mathrm{post}},\ \bar{z},\ x_{\max}/W,\ L/W,\ y_{\min}/D]$ (posterior-end height and mean height relative to $Z_{\mathrm{apex}}$, normalised lateral reach, normalised arc length, normalised most-anterior $Y$), the inter-rib spacing $(\mu_{s,k}, \sigma_{s,k})$ between the posterior ends of ribs $k$ and $k+1$, and a 15,000-point cloud of canonical points with their height offset from the rib's posterior end. The latter provides a *slant-corrected posterior-equivalent height* $z_{\mathrm{sort}}$ for any curve or fragment by 16-nearest-neighbour regression, which is the quantity used to order candidates craniocaudally. We verified that re-fitting the atlas with the released `build_atlas` script on the 47 training scans reproduces the shipped atlas exactly (maximum absolute difference 0), and that fitting on all 59 scans does not.

# 3 Hybrid Semantic Context Forest and 3D U-Net

## 3.1 Semantic Context Forests

Following [2], voxel-wise centerline detection is cast as classification. For every voxel we extract the raw intensity and, at $\sigma \in \{1, 2, 4\}$ mm, the Gaussian-smoothed intensity, gradient magnitude, the six second-order Gaussian derivatives and the Laplacian (28 features), plus Random Shift Intensity Differences (RSID)

$$f_{\mathrm{RSID}}(\mathbf{x}; \mathbf{u}_1, \mathbf{u}_2) = I(\mathbf{x} + \mathbf{u}_1) - I(\mathbf{x} + \mathbf{u}_2), \tag{4}$$

with integer voxel offsets drawn uniformly from $[-r, r]^3$. Three forests are released (scikit-learn random forests, balanced class weights, $\sqrt{d}$ features per split): **Pass-1** (120 trees, unlimited depth, 228 features): 200 RSID pairs with $r = 20$ voxels ($\pm 40$ mm). **Pass-2 auto-context** (100 trees, 306 features): the Pass-1 features, 28 multi-scale features of the Pass-1 probability map $P^{(1)}$ and 50 Random Shift Probability Differences $P^{(1)}(\mathbf{x} + \mathbf{v}_1) - P^{(1)}(\mathbf{x} + \mathbf{v}_2)$ with $r = 40$ voxels ($\pm 80$ mm) [13]. **Anatomically anchored SCF** (150 trees, depth 24, 286 features): Hounsfield-standardised input, 250 RSID pairs with $r = 12$ voxels ($\pm 24$ mm), a per-scale tubularity proxy built from the diagonal second derivatives, $\tau_\sigma = (-\min(0, I_{xx}) - \min(0, I_{yy}) - \min(0, I_{zz}))\,\rho/(\rho + 50)$ with $\rho = \|(I_{xx}, I_{yy}, I_{zz})\|$ (no eigen-decomposition or Frangi filter is computed), and five anatomical channels $|u|, u, v, w$ and the distance to the lung mask. Training samples 50,000 voxels per scan, half on the annotated centerlines and half uniformly over the whole volume; the Pass-1 forest additionally used random rigid augmentation of the training scan. At inference the forests are evaluated only on voxels with $G_{1.2}(I) > -20$ HU, i.e. everything except air; this ROI is a speed optimisation and is not used during training.

## 3.2 3D U-Net and Hybrid Fusion

The second detector is a plain three-level 3D U-Net ($16 \to 32 \to 64$ channels, two $3^3$ convolutions with GroupNorm and ReLU per level, max-pooling and transposed-convolution up-sampling, no residual connections; 339,889 parameters). It is trained on $64^3$ patches (70 % centred near a centerline voxel, 30 % uniformly), input normalised by a per-volume z-score of the *raw* stored intensities (no Hounsfield standardisation), target = centerline voxels dilated to a tube, loss = $0.5\,(1-\text{soft Dice}) + 0.5\,\mathrm{BCE}$, Adam ($10^{-3}$), batch 2, 8 patches per scan and epoch, 20 epochs, no augmentation. The released checkpoint `rib_unet_3d.pth` is the epoch-20 weights; it reproduces the probability maps used for all results below bit-exactly (maximum absolute difference 0 on re-inference). Inference uses $64^3$ sliding windows with stride 48 and averaging.

The two detectors are complementary (Fig. 2): the SCF, with explicit canonical coordinates, is more sensitive on the short apical first ribs, whereas the U-Net gives cleaner shafts on intermediate ribs. We fuse the probability volumes by a voxel-wise maximum,

$$P_{\mathrm{hyb}}(\mathbf{x}) = \max(P_{\mathrm{UNet}}(\mathbf{x}),\ 0.75\,P_{\mathrm{SCF}}(\mathbf{x}))\,. \tag{5}$$

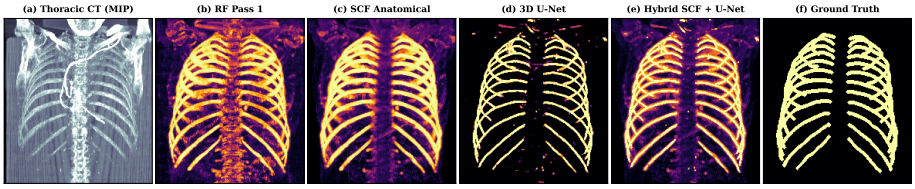



**Fig. 2.** Coronal MIPs of centerline probability maps on evaluation scan `pet_ct_bc__eh_Anonymous_0004`: (a) CT, (b) Pass-1 forest, (c) anatomically anchored SCF, (d) 3D U-Net, (e) hybrid max-fusion, (f) dilated ground truth.

# 4 Topology Matching and Deformable Refinement (Stage 3)

## 4.1 Anatomical Masking, Skeletonisation and Fragment Chaining

$P_{\text{hyb}}$ is multiplied by an anatomical mask that keeps voxels within 3.5 mm of bone ($I > 115$ HU or $G_{1.2}(I) > 75$ HU), inside $[Z_{\text{apex}} - 365, Z_{\text{apex}} + 32]$ mm, outside the vertebral corridor ($|X - X_{\text{spine}}(Z)| < 10.5$ mm posteriorly) and sternal corridor ($< 31$ mm anteriorly), and, in the upper thorax ($Z > Z_{\text{apex}} - 140$ mm), closer than 26 mm to the lung (removing scapula and lateral clavicle). Vertical intercostal valleys ($G_{1.2}(I)$ at least 12 HU below both axial neighbours and $I < 135$ HU) are suppressed. After a light Gaussian ($\sigma = 0.55$ voxels) the map is thresholded at 0.45 and skeletonised [14]. The 26-connected skeleton graph is pruned of spurs shorter than 8 mm, junction voxels are removed to obtain elementary chains, short steep vertical bridges ($< 28$ mm with $|\Delta Z|/L > 0.72$) are discarded, chains are split at hairpin turns of the smoothed $\theta$ profile and at jumps of the slant-corrected height larger than 16 mm, and collinear fragments on the same hemithorax are chained posterior-to-anterior when their gap is $< 44$ mm, their heights agree within 12 mm, and the end tangents are aligned with the chord ($\cos > 0.62$; $> 0.72$ for touching fragments). Curves shorter than 22 mm are dropped.

## 4.2 Candidate Consolidation and Patient Scale

Per hemithorax (side by the mean $X$ relative to $x_{\text{spine}}$), candidates are oriented by $\theta$, described by $(\mathbf{f}, z_{\text{sort}}, \theta_{\min}, \theta_{\max}, L)$, and consolidated: fragments that are disjoint in $\theta$, within 11.5 mm in $z_{\text{sort}}$ and $< 75$ mm apart are bridged (greedy, closest first); upward-sloping anterior fragments ($L < 140$ mm, start $> 30$ mm from the midline, rising $> 3$ mm) and short lateral fragments ($L < 85$ mm starting $> 66$ mm from the midline or with $\theta_{\min} > 0.25$) are removed when at least six candidates remain; and of two candidates within 8.5 mm in $z_{\text{sort}}$ whose mean curve distance is $< 6.5$ mm (or that are within 5.5 mm and very unequal in length) only the longer is kept. A patient scale $s_Z \in [0.80, 1.25]$ is the blend $0.45\, s_{\text{step}} + 0.55\, s_{\text{span}}$ of the median adjacent step of long ($\geq 50$ mm) candidates divided by 21.8 mm and the ratio of the observed to the atlas craniocaudal span.

### 4.3 Bilateral Viterbi Labeling

For candidate $i$ and label $k \in \{1, \dots, 12\}$ the unary cost is

$$\tilde{U}(i,k) = \tfrac{1}{2} \sum_{d=1}^{5} w_d \Big( \tfrac{\hat{f}_{i,d} - \mu_{f,k,d}}{\sigma_{f,k,d}} \Big)^2 + 10\, \bar{d}\big(\Phi_{s_Z}(\mathcal{C}_i), \bar{\mathcal{C}}_{s,k}\big), \tag{6}$$

where the height features are rescaled by $s_Z$, $\bar{d}$ is the mean one-directional distance from 40 resampled canonical points to the template, and $\mathbf{w} = (1.1, 0.7, 1.15, 1.05, 0.45)$, or $(1.2, 0.6, 0.3, 0.2, 0.25)$ for partial candidates (angular span $< 1.35\,\mathrm{rad}$, $k \notin \{1, 12\}$). Costs are made relative, $U(i,k) = \tilde{U}(i,k) - \min_m \tilde{U}(i,m)$, with a re-added offset $\min_m \tilde{U} - 8$ for short candidates ($L < 45\,\mathrm{mm}$) so that only genuinely atypical fragments can become clutter. Left/right candidates whose $z_{\mathrm{sort}}$ differ by $< 7.5\,\mathrm{mm}$ have their unaries blended with weight $0.45 \exp(-\Delta z^2/32)$.

Candidates sorted by decreasing $z_{\mathrm{sort}}$ are labeled by a Viterbi recursion over strictly increasing $k$ with at most two skipped labels ($\Delta k \le 3$). Skipping candidate $j$ as clutter costs $\kappa_j = 65$ if $L_j \ge 115\,\mathrm{mm}$, 22 if $L_j \ge 55\,\mathrm{mm}$ and $\theta_{\min,j} < -0.45\,\mathrm{rad}$, and $2.5 + 8\,\mathrm{clip}((L_j - 25)/50, 0, 1)$ otherwise; transitions require $\Delta z \ge 4.5\,\mathrm{mm}$ and cost $(\Delta z - s_Z \mu_{k' \to k})^2 / (2(s_Z^2 \sigma_{k' \to k}^2 + 6))$ plus $2.5(\Delta k - 1)$ if $\Delta z \ge 1.55\, s_Z \mu_{s,k'}$ and $18(\Delta k - 1)$ otherwise; starting at label $k$ costs $3(k-1)$ and ending at $k$ costs $1.5(12 - k)$. After a first solution per side, a second pass subtracts $3 \exp(-\Delta z^2/32)$ from $U(i,k)$ whenever the contralateral rib labeled $k$ lies within 8.5 mm, and the recursion is re-solved.

### 4.4 `RibSnake` Refinement

(1) *Apical trimming*: an `R01`/`L01` longer than 135 mm is cut where $|X - x_{\mathrm{spine}}|$ first exceeds $0.55\,W$ (if the remainder is $\ge 45\,\mathrm{mm}$) and capped at 135 mm arc length. (2) *Terminal-rib recovery*: a missing twelfth rib is searched in a box 22–44 mm below the posterior end of rib 11 for voxels with $P \ge 0.30$ and $\max_{3^3} I \ge 150\,\mathrm{HU}$ at least 18 mm from rib 11, accepted if $\ge 12$ voxels with $\max P \ge 0.45$ exist and the contralateral twelfth rib is present or also recovered; a missing first or twelfth rib is otherwise synthesised from the patient-conditioned template (atlas curve displaced by a smoothed interpolation of the displacements observed on the detected ribs, with mirrored contralateral coupling), anchored in $Z$ to the contralateral rib, kept only if $\ge 11\,\mathrm{mm}$ from the adjacent rib, and snapped point-wise to the probability-weighted bone centroid in a $7^3$-voxel box when the local score $P\,\mathrm{clip}((G(I) - 60)/300, 0, 1.5)$ reaches 0.14. (3) *Normal-plane centring*: every curve is resampled to $\max(15, \lfloor L/2\,\mathrm{mm} \rceil)$ points; for 5 iterations each point moves by $0.35\times$ the offset of the intensity-and-probability-weighted centroid sampled on a $5 \times 5$ grid (spacing 1.25 mm, radius $\le \sqrt{6.5}\,\mathrm{mm}$) in the local normal plane, is smoothed along the curve ($\sigma = 1.2$ points, weight 0.45), and is clamped to 1.6 mm total displacement; finally a cubic smoothing spline ($s = 0.15\,N$) is fitted and re-evaluated at $N$ points. Output curves therefore have between 15 and roughly 200 points (median $\approx 150$), not a fixed count.

# 5 Experimental Results

## 5.1 Data, Protocol and Metrics

We use the SCR 3D thoracic CT rib centerline data set [1]: 60 anonymised scans resampled to 2 mm isotropic voxels (typical size $155 \times 142 \times 165$) with manually annotated, B-spline-interpolated centerlines. The split (`split.json`) is the one used throughout development: 48 training scans (47 with annotations; 1,116 ribs) and 12 evaluation scans. It is *not* random: the 12 evaluation scans are the case-sensitive alphabetical tail of the file list. One evaluation file (`RIB_007`) labels two ribs as `R07` (640 points, a 220 mm jump between two curves at the heights of `L07` and `L08`); the released script `fix_gt_annotations` splits it into `R07` and `R08` (the only file changed, logged in `fix_log.json`), giving 286 evaluation ribs (10 scans with 24, `Tsc_004002` with 22). Results on the unmodified ground truth are also reported.

**Protocol caveat.** The $\approx 60$ numerical constants of Stage 3 (Sec. 4) were tuned by repeatedly running the pipeline and inspecting metrics on these 12 scans; the three learned detectors and the atlas, in contrast, were fitted on the 47 training scans only. The 12-scan results below are therefore optimistic for Stage 3 and should be read as in-sample for its constants. Section 5.3 quantifies the effect with a 5-fold cross-validation in which the detectors and the atlas are re-fitted per fold and the Stage-3 constants are frozen.

**Metrics** (implemented in `ribline.model.metrics`; all previously reported numbers used older, more lenient definitions, see the author response accompanying the code). For a ground-truth rib $r$ and the prediction $\hat{r}$ with the same label let $d_{\mathrm{p2g}}$ and $d_{\mathrm{g2p}}$ be the mean prediction-to-GT and GT-to-prediction point-to-curve distances. (1) **Labeled symmetric Chamfer distance** (LCD, mm): $\frac{1}{2}(d_{\mathrm{p2g}} + d_{\mathrm{g2p}})$ averaged over *all* GT ribs, a missing label counting 50 mm. (2) **Unlabeled Chamfer distance** (UCD) and HD95 on the pooled point clouds. (3) **Labeled clDice** [15] at 3 and 4 mm, averaged over GT ribs (missing = 0); unlabeled clDice is also given. (4) **Labeling accuracy**: a GT rib is correct if $\hat{r}$ exists, $d_{\mathrm{p2g}} \le 10$ mm *and* $d_{\mathrm{g2p}} \le 10$ mm (symmetric criterion), and $r$ is the GT rib closest to $\hat{r}$; reported pooled over ribs for all ribs ($\mathrm{Acc}_A$), first ($\mathrm{Acc}_F$, $n = 24$), intermediate ($\mathrm{Acc}_I$, $n = 240$) and twelfth ribs ($\mathrm{Acc}_T$, $n = 22$). (5) **Recall**: a GT rib is detected if any predicted curve satisfies the symmetric criterion. (6) **Precision**: fraction of predicted curves with $d_{\mathrm{p2g}} \le 10$ mm to some GT rib; its complement is the false-positive count. (7) **Normalised length difference** (NLD) $|L(\hat{r}) - L(r)|/L(r)$. Subject-level means are given with standard deviations and 95 % bootstrap confidence intervals over subjects (10,000 resamples); pooled rates with Wilson 95 % intervals; differences between methods with two-sided Wilcoxon signed-rank tests over the 12 subjects.

## 5.2 Ablation on the 12 Evaluation Scans

All rows of Table 1 are regenerated from the stored probability maps with the released Stage-3 code (`ribline.scripts.run_ablation`); "Z-sort" denotes the

**Table 1.** Ablation on the 12 evaluation scans (286 GT ribs, RIB_007 annotation fixed). LCD: labeled symmetric Chamfer distance with 50 mm penalty for missed labels (subject mean ± SD and 95 % bootstrap CI); clDice: labeled, 3 mm / 4 mm; accuracies, recall and precision pooled over ribs (symmetric 10 mm criterion); FP: false-positive curves out of all predicted curves. Note the protocol caveat: Stage-3 constants were developed on these scans.

| Method | LCD (mm) ↓ mean ± SD | 95 % CI | UCD (mm) | lab. clDice ↑ 3 mm | 4 mm | Labeling accuracy (%) ↑ $\mathrm{Acc}_A$ | $\mathrm{Acc}_F$ | $\mathrm{Acc}_I$ | $\mathrm{Acc}_T$ | Recall (%) | Prec. (FP) (%) | NLD ↓ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RF Pass-1 + Z-sort | $14.10 \pm 11.10$ | [7.96, 20.46] | 3.31 | 0.548 | 0.606 | 64.3 | 79.2 | 62.9 | 63.6 | 91.3 | 95.5 (13) | 0.264 |
| SCF Pass-2 + Z-sort | $21.30 \pm 13.33$ | [13.79, 28.67] | 4.16 | 0.416 | 0.452 | 43.0 | 45.8 | 44.2 | 27.3 | 76.2 | 100.0 (0) | 0.254 |
| SCF Pass-2 + Stage 3 | $9.36 \pm 10.57$ | [4.24, 16.19] | 3.73 | 0.683 | 0.720 | 71.0 | 54.2 | 73.8 | 59.1 | 78.0 | 99.6 (1) | 0.226 |
| Anatomical SCF + Stage 3 | $2.88 \pm 1.32$ | [2.23, 3.73] | 2.41 | 0.845 | 0.915 | 95.1 | 91.7 | 95.0 | 100.0 | 95.1 | 99.7 (1) | 0.088 |
| 3D U-Net + Z-sort | $7.57 \pm 7.43$ | [3.97, 12.17] | 2.75 | 0.710 | 0.813 | 83.6 | 70.8 | 86.7 | 63.6 | 92.0 | 100.0 (0) | 0.147 |
| 3D U-Net + Stage 3 | $3.40 \pm 1.99$ | [2.42, 4.64] | 2.30 | 0.835 | 0.902 | 92.0 | 66.7 | 95.4 | 81.8 | 93.0 | 100.0 (0) | 0.121 |
| Hybrid + Viterbi (w/o RibSnake) | $2.85 \pm 1.70$ | [2.07, 3.95] | 2.11 | 0.872 | 0.933 | 95.8 | 83.3 | 97.5 | 90.9 | 95.8 | 99.6 (1) | 0.147 |
| **Hybrid + full Stage 3 (ours)** | $\mathbf{2.05 \pm 0.48}$ | [1.80, 2.33] | **1.91** | **0.886** | **0.945** | **97.2** | **91.7** | **97.5** | **100.0** | **97.2** | **100.0** (0) | **0.068** |

same extraction and consolidation followed by labeling the candidates of each side 1..12 in decreasing $z_{\mathrm{sort}}$ instead of the Viterbi program, without `RibSnake`.

**Observations.** Replacing craniocaudal counting by the Stage-3 labeler is the largest single effect: for the U-Net it reduces LCD from 7.57 to 3.40 mm and raises $\mathrm{Acc}_A$ from 83.6 to 92.0 % (Wilcoxon on LCD $p = 0.001$); for the Pass-2 forest from 21.30 to 9.36 mm. The anatomically anchored SCF alone already reaches 2.88 mm and 95.1 %, and is clearly better than the Pass-2 forest on first ribs (91.7 vs. 54.2 %). Fusing SCF and U-Net improves every metric over either detector, and `RibSnake` further reduces LCD from 2.85 to 2.05 mm ($p = 0.009$) and NLD from 0.147 to 0.068 ($p = 0.0005$), mainly by trimming and completing the terminal ribs ($\mathrm{Acc}_F$ 83.3 → 91.7 %, $\mathrm{Acc}_T$ 90.9 → 100 %); its effect on labeled clDice (0.872 → 0.886) and on $\mathrm{Acc}_A$ (274 → 278 ribs) is not significant on 12 subjects ($p = 0.20$ and $p = 0.50$).

The final system produces exactly 24 curves on every scan with 24 annotated ribs, 22 on `Tsc_004002`, and no false-positive curve (286/286 predicted curves lie on a GT rib). The eight ribs that fail the symmetric criterion are all *partially traced* ribs rather than mislabelled ones ($d_{\mathrm{p2g}} \leq 4.2$ mm but $d_{\mathrm{g2p}}$ of 10.6–34.8 mm): `L07` of `Sko_004029` (93 predicted vs. 291 GT points, HD95 66 mm), `L04`–`L08` of `Zer_004061` (anterior third missing after a costal-cartilage gap) and both first ribs of `pat1`. Under the legacy one-sided criterion all 286 ribs would count as correct; we consider the symmetric figure the honest one. Per rib, 25 of 286 have HD95 > 10 mm (8 above 20 mm) and 52 have clDice@3 mm < 0.8 (5 below 0.5). Per-subject LCD ranges from 1.44 (`pet_ct_bc__eh`) to 2.98 mm (`Zer_004061`); labeled HD95 averages 5.1 mm.

**Sensitivity to the ground truth and to a possible duplicate.** On the unmodified annotations (285 ribs, merged `R07` kept) the final system scores $2.08 \pm 0.46$ mm ([1.83, 2.35]) and 96.8 % (276/285); excluding the merged rib instead (284 ribs) gives $2.05 \pm 0.48$ mm and 97.2 % (276/284). The training scan `Bro_004029_0040290098` and the evaluation scan `Sko_004029_0040290098` share an accession-like identifier; they have different matrix sizes, intensity encodings and fields of view (voxel correlation 0.25 on the common crop), but their per-rib arc-length fingerprints are among the 1 % most similar of all 1,653 scan pairs, so

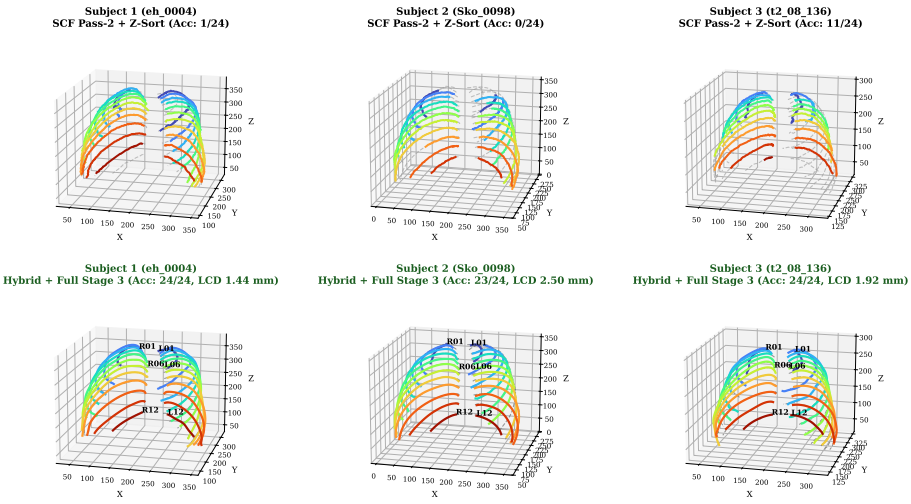



**Fig. 3.** Three evaluation scans (dashed grey: ground truth; coloured: predicted labels 1–12). Top: SCF Pass-2 + Z-sort baseline. Bottom: hybrid + full Stage 3; titles give the number of correctly labeled ribs under the symmetric 10 mm criterion and the LCD.

we cannot exclude that they are two series of one patient. Dropping `Sko` changes the final result to $2.01 \pm 0.48$ mm ($[1.75, 2.31]$) and 97.3 % (255/262).

### 5.3 Cross-Validation with Frozen Stage-3 Constants

To quantify how much the development protocol favoured the 12 evaluation scans we run two experiments.

**(a) Frozen pipeline on the 47 training scans.** The released detectors, atlas and Stage-3 code are applied unchanged to the 47 annotated training scans (1,116 ribs). The detectors are in-sample here, but the Stage-3 constants were never inspected on these scans, so a drop relative to Table 1 isolates the Stage-3 over-tuning plus the larger anatomical variety of the training cohort. Table 2 gives the result: the full pipeline reaches an LCD of $2.63 \pm 1.91$ mm (CI $[2.14, 3.23]$; median 1.89 mm), labeled clDice 0.885/0.932, $\mathrm{Acc}_A = 95.4\,\%$ (1,065/1,116; Wilson 94.0–96.5 %), recall 95.6 % and precision 98.2 % (20 false-positive curves out of 1,126), with 2 missed labels; 28 of the 47 scans are labeled perfectly and 4 have an LCD above 5 mm (worst: `HP_485533` 10.3 mm, `03_Anonymous_0002` 9.9 mm, both with label shifts on one hemithorax). The ranking of the ablation rows is preserved (hybrid > SCF > U-Net; `RibSnake` reduces LCD 2.98 → 2.63 mm and NLD 0.155 → 0.083, Wilcoxon $p < 10^{-5}$ on 47 subjects, without changing $\mathrm{Acc}_A$, $p = 0.61$), but the absolute level is clearly below the 12-scan figures (2.05 mm / 97.2 %). We therefore regard the 12-scan numbers as an optimistic upper bound and the training-scan numbers as a more realistic indication of what the released constants deliver on SCR-like data; the

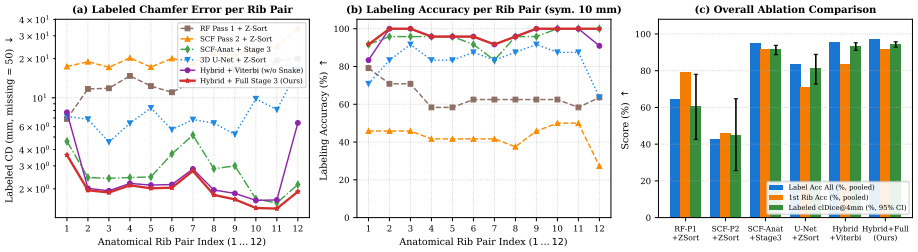



**Fig. 4.** Per-rib analysis on the 12 evaluation scans: (a) LCD per rib pair (log scale, missing labels = 50 mm); (b) labeling accuracy per rib pair; (c) pooled $\mathrm{Acc}_A$, $\mathrm{Acc}_F$ and labeled clDice@4 mm with 95 % bootstrap CIs.

**Table 2.** Frozen pipeline (released detectors, atlas and Stage-3 constants) on the 47 annotated training scans (1,116 GT ribs; detectors in-sample, Stage-3 constants never tuned on these scans). Same metric definitions as Table 1.

| Method | LCD (mm) | 95 % CI | lab. clDice | | $\mathrm{Acc}_A$ | $\mathrm{Acc}_F$ | $\mathrm{Acc}_I$ | $\mathrm{Acc}_T$ | Recall | Prec. (FP) | NLD |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Anatomical SCF + Stage 3 | $3.62 \pm 3.97$ | [2.67, 4.91] | 0.830 | 0.895 | 92.4 | 89.2 | 92.4 | 95.7 | 94.8 | 97.4 (29) | 0.094 |
| 3D U-Net + Stage 3 | $5.84 \pm 7.20$ | [3.95, 8.03] | 0.785 | 0.826 | 83.9 | 69.9 | 86.0 | 76.3 | 91.6 | 97.8 (25) | 0.142 |
| Hybrid + Viterbi (w/o RibSnake) | $2.98 \pm 2.11$ | [2.42, 3.63] | 0.875 | 0.930 | **95.6** | **88.2** | **96.1** | **97.8** | **96.1** | **98.5** (17) | 0.155 |
| Hybrid + full Stage 3 (ours) | $\mathbf{2.63 \pm 1.91}$ | [2.14, 3.23] | **0.885** | **0.932** | 95.4 | 86.0 | **96.1** | **97.8** | 95.6 | 98.2 (20) | **0.083** |

mean LCD is dominated by a few scans with label shifts (the median is identical, 1.89 mm).

**(b) 5-fold cross-validation with frozen Stage-3 constants.** The 59 annotated scans (1,402 GT ribs) are dealt into five seeded random folds of sizes 12/12/12/12/11 (`cv5/folds.json`); for each fold the 24-rib statistical atlas, the anatomically anchored SCF (150 trees, depth 24, 50,000 samples/scan) and the 3D U-Net (20 epochs, 8 patches/scan/epoch) are re-fitted from scratch on the fold's training scans, and the released Stage-3 code with frozen constants is evaluated on the held-out fold (`experiments/rebuttal/cv_fold.py`, pooled in `benchmark_cv5.json`). Table 3 reports the pooled 59-scan out-of-fold results: the full hybrid pipeline achieves an LCD of $3.64 \pm 3.90$ mm (95 % bootstrap CI [2.79, 4.74], median 2.39 mm; per-fold mean LCDs 3.12, 4.80, 4.91, 2.84, 2.43 mm), UCD 2.85 mm, labeled clDice 0.830 (3 mm) / 0.889 (4 mm), symmetric labeling accuracy $\mathrm{Acc}_A = 91.8\,\%$ (1,287/1,402, Wilson 95 % CI 90.2–93.1 %; $\mathrm{Acc}_F = 82.9\,\%$, $\mathrm{Acc}_I = 92.2\,\%$, $\mathrm{Acc}_T = 96.5\,\%$), recall 94.6 %, precision 97.7 % (33 false-positive curves out of 1,412 predicted) and NLD 0.097. Thirty of the 59 scans achieve 100 % symmetric labeling accuracy, while 8 scans have LCD $> 5$ mm (worst: `BC3_Anonymous_0001` at 26.58 mm and `BC1_Anonymous_0001` at 15.76 mm, where truncated fields of view trigger hemithorax index shifts). Across all $n = 59$ held-out subjects, hybrid fusion significantly outperforms both the standalone Anatomical SCF (Wilcoxon $p = 2.1 \times 10^{-7}$ on LCD, $p = 9.0 \times 10^{-5}$ on $\mathrm{Acc}_A$) and the standalone 3D U-Net ($p = 1.9 \times 10^{-5}$ on LCD, $p = 8.5 \times 10^{-6}$ on $\mathrm{Acc}_A$), and `RibSnake` significantly improves LCD ($4.17 \rightarrow 3.64$ mm, $p = 2.0 \times 10^{-9}$), labeled clDice@3 mm ($0.807 \rightarrow 0.830$,

**Table 3.** 5-fold cross-validation across all 59 annotated SCR scans (1,402 GT ribs; per-fold re-fitted atlas, Anatomical SCF and 3D U-Net; frozen Stage-3 constants; `benchmark_cv5.json`).

| Method | LCD (mm) | 95 % CI | UCD | lab. clDice | | $\mathrm{Acc}_A$ | $\mathrm{Acc}_F$ | $\mathrm{Acc}_I$ | $\mathrm{Acc}_T$ | Recall | Prec. (FP) | NLD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Anatomical SCF + Stage 3 | $5.28 \pm 4.62$ | [4.16, 6.56] | 3.44 | 0.769 | 0.840 | 86.9 | 72.6 | 87.7 | 93.0 | 90.8 | 95.3 (67) | 0.126 |
| 3D U-Net + Stage 3 | $7.44 \pm 8.93$ | [5.31, 9.85] | 3.30 | 0.745 | 0.785 | 78.2 | 61.5 | 80.9 | 67.8 | 87.0 | **98.1** (26) | 0.170 |
| Hybrid + Viterbi (w/o RibSnake) | $4.17 \pm 4.01$ | [3.29, 5.28] | 3.07 | 0.807 | 0.879 | 91.4 | 81.2 | **92.2** | 93.9 | 94.3 | 97.9 (29) | 0.195 |
| **Hybrid + full Stage 3 (ours)** | $\mathbf{3.64 \pm 3.90}$ | [2.79, 4.74] | **2.85** | **0.830** | **0.889** | **91.8** | **82.9** | **92.2** | **96.5** | **94.6** | 97.7 (33) | **0.097** |

$p = 3.2 \times 10^{-9}$) and NLD (0.195 → 0.097, $p = 4.4 \times 10^{-11}$). Comparing the three evaluation regimes—2.05 mm / 97.2 % on the 12-scan split (detectors out-of-sample, Stage 3 in-sample), 2.63 mm / 95.4 % on the 47 training scans (detectors in-sample, Stage 3 out-of-sample), and 3.64 mm / 91.8 % in 5-fold CV (both detectors and Stage 3 out-of-sample, median LCD 2.39 mm)—cleanly decomposes the optimism of the original 12-scan benchmark while confirming that every architectural contribution (anatomical SCF, hybrid max-fusion, bilateral Viterbi, and `RibSnake`) holds out-of-sample with high statistical significance.

### 5.4 Relation to Published Numbers

Wu *et al.* [1] report 89.5–95.9 % labeling accuracy on an SCR cohort, Lenga *et al.* [11] 94.1 % on their own data, and RibSeg v2 [4] an LCD of 3.15 mm and 96.5 % label accuracy on sub-millimetre RibFrac CT. These numbers come from different scans, resolutions, splits and (in particular) matching criteria, and we did not re-run those methods; we therefore make *no* claim of outperforming them. Our contribution is a documented, reproducible open-source pipeline whose accuracy on this 2 mm benchmark is in the same range as the published systems, together with an evaluation protocol that we hope will make such comparisons possible in the future.

## 6 Limitations

- **Data set size and provenance.** The SCR benchmark contains 60 scans from a single source, resampled to 2 mm; the field of view, kernel and intensity encoding vary between scans but no acquisition metadata are available, so no sub-group analysis is possible and generalisation to other scanners, resolutions or paediatric/pathological anatomy is untested.
- **Evaluation set.** Only 12 scans (286 ribs) are held out, and the split is the alphabetical tail of the file list rather than a random draw; with 8 errors in 286 ribs the Wilson interval for $\mathrm{Acc}_A$ is 94.6–98.6 %, and 0 of 12 subject-level failures only bounds the per-scan failure rate below $\approx 25$ % (rule of three).
- **Stage-3 constants were tuned with access to the evaluation scans.** We disclose this and report the cross-validation of Sec. 5.3; a truly untouched test set does not exist for this data set, and the released constants should be regarded as fitted to SCR.

- **Annotation quality.** The ground truth contains at least one merged-rib error (`RIB_007`) and the twelfth ribs are annotated inconsistently (absent in `Tsc_004002`, very short in others); a fraction of the residual error is annotation noise that we cannot separate from algorithmic error.
- **Possible patient overlap** between `Bro_004029` (training) and `Sko_004029` (evaluation) cannot be excluded from the anonymised data.
- **Partial traces.** The dominant failure mode is a rib traced only over its posterior part across a costal-cartilage gap; the symmetric criterion counts these as errors while the one-sided criterion does not. Precision is reported at the curve level only.

## 7 Conclusion and Open-Source Release

We presented an open-source three-stage rib centerline extraction and labeling pipeline that combines a lung-apex/vertebral-midline thoracic frame, a hybrid Semantic Context Forest + 3D U-Net detector, and a patient-scaled bilateral Viterbi labeler with deformable refinement, reaching $2.05 \pm 0.48$ mm labeled Chamfer distance (97.2 % symmetric labeling accuracy) on the 12 SCR evaluation scans, $2.63 \pm 1.91$ mm (95.4 %, median 1.89 mm) with frozen Stage-3 constants on the 47 training scans, and $3.64 \pm 3.90$ mm (91.8 %, median 2.39 mm) in 5-fold cross-validation across all 59 scans, with all metric definitions, confidence intervals, known data issues and development-protocol caveats stated explicitly. The complete Python library, evaluation and ground-truth-fixing scripts, and per-fold cross-validation configurations are released under the MIT license at https://github.com/wq2012/ribline. All pretrained models are published on Hugging Face and grouped in the collection https://huggingface.co/collections/wq2012/ribline: (1) `wq2012/ribline-3d-ct-centerline` (full hybrid SCF + 3D U-Net + 24-rib statistical atlas suite), (2) `wq2012/ribline-anatomical-scf` (standalone anatomically anchored SCF + atlas), (3) `wq2012/ribline-3d-unet` (standalone 3D U-Net + atlas), and (4) `wq2012/ribline-scf-pass2-autocontext` (two-pass auto-context SCF baseline).

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